

Dr. Yannis C. Yortsos,
Editor in-chief, PNAS Nexus

September 2, 2026

Dear Dr. Yortsos,

We are pleased to submit our manuscript, “**Regulatory stochasticity drives opposing phenotypic outcomes in cell-fate decision networks**” by Hari, Gupta, Shivakumar, Kulkarni, Salgia, Jolly and Levine for consideration of publication in PNAS Nexus as a research report.

Models of gene regulatory networks underlying cell-fate decisions typically assume that regulatory interaction strengths are fixed, even though transcription-factor efficacy can vary over time due to changes in molecular conformation, post-translational modification, cofactor availability, and chromatin accessibility. Here, we ask how temporal fluctuations in regulatory strength themselves reshape the phenotypic landscapes generated by multistable regulatory networks.

Using ensembles of mutually inhibitory circuits together with larger Boolean models of epithelial–mesenchymal plasticity and gonadal-fate determination, we show that distinct modes of regulatory stochasticity can drive qualitatively opposing cell-fate outcomes. Additive fluctuations favor less differentiated hybrid states, whereas multiplicative fluctuations that progressively enhance regulatory efficacy favor more committed, single-high states. Importantly, fixed-parameter controls sampled from the same noise-generated parameter distributions do not fully reproduce these effects, demonstrating that the temporal evolution of regulatory interactions contributes information beyond static parameter heterogeneity.

We believe this work will be of broad interest to readers studying gene regulatory networks, stochastic biological dynamics, cell-fate determination, and non-genetic phenotypic heterogeneity. More generally, our results suggest that the temporal statistics of regulatory efficacy can themselves act as a determinant of cellular phenotype, providing a framework for connecting molecular sources of regulatory variability to developmental and pathological cell-state transitions.

Thank you for considering our manuscript for publication in PNAS Nexus.

Sincerely,

Kishore Hari
on behalf of the authors